

Haseeb Manzoor

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ABOUT ME

AI Engineer with deep expertise in Generative AI, Large Language Models (LLMs), and MLOps. Skilled in building Transformer architectures and large language models(LLM) from scratch, fine-tuning state-of-the-art models, and developing agentic AI systems using LangChain and LangGraph. Experienced in designing end-to-end ML pipelines, CI/CD automation, and cloud deployment on AWS.

EDUCATION

National University of Sciences and Technology (NUST) <i>Master of Science in Artificial Intelligence</i>	2025 – present
COMSATS University Islamabad, Sahiwal Campus <i>Bachelor of Science in Electrical Engineering</i>	2020 – 2024
Army Public School and College System, Okara Cantt <i>FSc Pre-Engineering</i>	2018 – 2020

TECHNICAL SKILLS

Languages: Python
Technologies: TensorFlow, PyTorch, Scikit-learn, Hugging Face, Advanced Transformer Architectures, LangChain, LangGraph, LangSmith, FineTuning, FastAPI, Streamlit
MLOps: Docker, MLflow, Git, GitHub Actions (CI/CD), DVC, AWS (EC2, S3, ECR, IAM)
Databases: MySQL (basic), Weaviate, ChromaDB, PineCone

PROJECTS

Encoder–Decoder Transformer(En–Ur Translation) <i>PyTorch</i>	2025
– Designed, implemented, and trained a complete Encoder–Decoder Transformer architecture from scratch. – Implemented sinusoidal positional encoding, multi-head attention, cross-attention, masked attention, and padding masking. – Used SwiGLU activation and RMSNorm with Pre-Normalization for improved stability and gradient flow. – Built a tokenizer using SentencePiece and trained on 53K parallel English–Urdu sentence pairs. – Implemented efficient checkpointing for resuming training and tracking best model performance. – Achieved BLEU score of 22.96 after 20 epochs (Train Loss = 1.5371, Validation Loss = 2.5357).	
AI Trip Organizer Agent <i>LangChain, LangGraph, Streamlit, FastAPI</i>	2025
– Built a modular trip planner with multi-step workflows and real-time interaction. – Integrated Google APIs, weather, currency, and expense modules. – Deployed with Streamlit (frontend) and FastAPI (backend) for real-time user interaction	
Vehicle Insurance Prediction System <i>FastAPI, AWS, Docker</i>	2025
– Developed a CI/CD pipeline using GitHub Actions and deployed the system on AWS EC2, with AWS S3 for model storage. – Used MongoDB Atlas for cloud-based data storage and containerized the application with Docker. – Built the backend using FastAPI and designed basic UI components with HTML and CSS.	
RAG-based Agentic Application <i>LangChain, LangGraph, Streamlit, FastAPI, Docker</i>	2025
– Developed a modular RAG-based agentic system with predefined domain-specific vector stores (Dermatology, Psychiatry, Legal) and support for custom PDF uploads. – Implemented hybrid retrieval (BM25 + dense embeddings) with coherence ranker, redundancy filter, long-context reordering, and contextual compression for improved answer accuracy. – Integrated SQLite checkpointer for chat persistence and conversation summary memory for Memory. – Enabled multimodal input (text + audio) and containerized the application with Docker Compose for local testing.	

DevelopersHub Corporation (Remote)

Mar 2025 – May 2025

AI/ML Engineer

- * Developed a deep learning sentiment analysis model using LSTM, achieving 90% accuracy on the IMDB dataset.
- * Developed a FinBERT-based sentiment analysis system with 93% accuracy on Financial review data.
- * Developed fake news detection pipeline using NLP + ML.
- * Time-series anomaly detection using LSTM + Isolation Forest on financial data.
- * Credit risk model with SMOTE + Random Forest with 92% accuracy.
- * BERT-based emotion recognition with optimized F1-score.